

Multiple Choice (10 marks)

1. The red glow in the neon tube of an advertising sign is a result of
 - (a) Refraction
 - (b) de-excitation
 - (c) Chemiluminescence
 - (d) Thermal radiation
 - (e) Phosphorescence
2. A coil of 60 ohms inductive reactance in series with a resistance of 25 ohms is connected to an a-c line of 130 volts. What will be the current through the circuit?
 - (a) 2.5 amp
 - (b) 0.5 amp
 - (c) 2.8 amp
 - (d) 3.0 amp
 - (e) 1.0 amp
3. If light of wavelength 635 nm causes a certain amount of Rayleigh scattering, light of what wavelength will give exactly 10 times as much scattering?
 - (a) 700 nm
 - (b) 450 nm
 - (c) 750 nm
 - (d) 300 nm
 - (e) 357 nm
4. At what velocity is the mass of a particle twice its rest mass?
 - (a) 3.2×10^{10} cm/sec
 - (b) 3.0×10^{10} cm/sec
 - (c) 2.8×10^{10} cm/sec
 - (d) 2.4×10^{10} cm/sec
 - (e) 1.8×10^{10} cm/sec
5. The pressure at the bottom of a pond does NOT depend on the
 - (a) salinity of the water.
 - (b) proximity to the equator.

- (c) presence of fish in the pond.
- (d) type of pond vegetation.
- (e) acceleration due to gravity.

True / False (10 marks)

Answer True or False

6. True or False: The Mars Exploration Rover Spirit is tilted towards the north to maximize solar power generation while avoiding a sand storm.
7. True or False: The reflectance for light of normal incidence at a water-glass boundary is 0.70% for crown glass and 1.20% for flint glass.
8. True or False: The frequency of the next higher harmonic for a closed organ pipe with a fundamental frequency of 131 Hz is 196 Hz.
9. True or False: An object can have zero acceleration if the net external forces are perfectly balanced, regardless of their individual magnitudes.
10. True or False: The impedance of a 1-henry inductor at angular frequencies of 100, 1000, and 10,000 rad/sec is 500 ohms, 5000 ohms, and 50 ohms, respectively.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. A 5-cm plane mirror can reflect the entire image of a tree when positioned 30 cm from an observer 100 m away. Calculate the height of the tree and discuss the principles involved in this scenario.
12. Discuss how operating a heat engine, such as a jet engine, at high temperatures influences its efficiency. Analyze the underlying thermodynamic principles at play.

End of Paper